

AI/ML-Enabled Adaptive Noise Cancellation (ANC) for Defence Applications

Hackathon Problem Statement — Analysis & Proposed Solution

Derived from: M. Ruhland, J. Bitzer, M. Brandt, S. Goetze — "Reduction of Gaussian, Supergaussian, and Impulsive Noise by Interpolation of the Binary Mask Residual," *IEEE/ACM TASLP*, Vol. 23, No. 10, Oct 2015

1. Hackathon Problem Statement

To develop an AI/ML-enabled adaptive noise cancellation (ANC) system that effectively suppresses stationary, non-stationary, and impulsive defence noises while maintaining high speech intelligibility and real-time performance on embedded hardware.

Key requirements extracted from the statement:

- Handle three noise categories: **stationary** (e.g. engine hum, HVAC), **non-stationary** (e.g. wind, moving vehicle rattle), and **impulsive** (e.g. gunfire, explosions).
- Preserve **speech intelligibility** — the system must not damage the voice/command signal while removing noise.
- Must be **adaptive** — implying an AI/ML component that adjusts to changing noise conditions.
- Must run in **real time** on **embedded hardware** — ruling out large, compute-heavy deep-learning-only pipelines.

2. The Underlying Problem (As Identified in the Research Paper)

2.1 Classical noise-reduction methods assume the wrong noise model

Most established denoising algorithms (e.g. the Wiener filter, Ephraim-Malah MMSE estimator) assume noise follows a simple, smooth **Gaussian** statistical distribution. However, real-world noise — car noise, rain, gunfire, old optical soundtrack degradation — is frequently **supergaussian**: spikier and more unpredictable than Gaussian noise. Applying a Gaussian-based tool to non-Gaussian noise causes measurable error and can even damage the underlying desired signal.

2.2 A neglected "in-between" noise category

Existing techniques split into two families, and both fail on a specific, important case:

- **Frequency-domain methods** (good for steady hiss) operate on the whole signal block, risking damage to the real signal.
- **Time-domain click-removal methods** work well only for rare, isolated single clicks (e.g. a scratch on a vinyl record) — not for a continuous stream of tiny impulses.
- Noise such as rain, applause, gunfire bursts, or old film soundtrack grain is technically impulsive at the micro level but sounds continuous at the macro level — it falls into neither category and is poorly handled by both approaches.

2.3 Existing "clean the whole spectrum" methods harm the desired signal

The paper's own baseline comparison (an established method called LSAR — Least Squares Autoregressive) demonstrates this concretely: at higher input signal quality, LSAR actually produced **negative** SNR improvement for pop music — it mistook real musical transients (e.g. drum hits) for noise spikes and wrongly "corrected" (damaged) them. This is the central risk for any noise-cancellation system: the inability to reliably distinguish *real sharp signal* from *sharp noise*.

2.4 No predictable control over the cleaning-vs-damage trade-off

Baseline methods like LSAR offer no reliable way to set their aggressiveness in advance — the outcome can only be observed after running the algorithm. There is no dial with known, predictable consequences for how much noise is removed versus how much signal is risked.

2.5 Aggressive cleaning introduces new artifacts

Even the theoretically "perfect" separation method (the Ideal Binary Mask) sounds worse to real listeners at high signal quality than the paper's proposed method, because removing all noise energy mathematically still leaves behind audible warbling artifacts known as "musical noise."

3. The Paper's Solution — Binary Mask Residual Interpolation (BMRI)

The paper proposes a lightweight, two-stage hybrid pipeline:

Stage 1 — Time-Frequency Binary Masking

The noisy signal is split, block by block, into a "target" (likely clean signal) and "residual" (likely noise) using a frequency-dependent, adaptively-tracked threshold (fast release / slow attack), so brief real transients are not mistaken for noise.

Stage 2 — Autoregressive (AR) Detection and Interpolation

Only the noisy "residual" portion is processed further: an AR model predicts what each sample should be from its neighbours; the worst $\gamma\%$ (a tunable **detection percentage**) of mismatching samples are flagged as impulsive and mathematically interpolated back in. A final spectral-correction pass removes leftover rumble artifacts before recombining with the target signal.

This design is deliberately lightweight — small block sizes, low AR model orders, a single interpolation pass, no large neural network — making it realistic for real-time, embedded operation, and it is proven (via objective SNR/perceptual measures and real listening tests) to outperform the standard baseline, especially for spiky/impulsive noise, while better preserving the desired signal.

4. Mapping the Paper to the Hackathon Problem Statement

Hackathon Requirement	What the Paper Already Provides
Impulsive noise (gunfire, explosions, mechanical clanks)	AR detection + interpolation stage — the paper's strongest, best-validated result
Non-stationary / spiky noise (sustained bursts, vehicle rattle)	Same AR stage — this is precisely what "supergaussian" noise means in the paper
Stationary noise (engine hum, HVAC)	Binary masking stage — present but the paper's own acknowledged weak point
Preserve speech intelligibility	Adaptive threshold design + spectral-correction step, built specifically to avoid damaging the desired signal
Real-time performance on embedded hardware	Small block sizes (2048 samples), low AR orders (16 / 32), single-pass processing — no heavy neural computation

5. Proposed Hackathon Solution

Rather than building a large end-to-end deep-learning denoiser (too heavy for embedded, real-time defence hardware), the proposal is to adopt the paper's proven lightweight architecture as the core engine, and add targeted AI/ML components exactly where the paper is weak or manual — turning a static, hand-tuned academic algorithm into a genuinely adaptive, defence-ready system.

5.1 Core Engine — Reuse the Two-Stage Architecture

Implement the binary-masking + AR detection/interpolation pipeline as the system backbone. This gives a working, low-latency base capable of handling stationary, non-stationary, and impulsive noise, without relying on a heavy neural model for the core signal path.

5.2 Innovation 1 — ML-Driven Adaptive Detection Percentage (γ)

The paper hand-tunes its detection percentage γ offline using an analytic kurtosis chart. Replace this with a small, fast real-time classifier (SVM / decision tree / tiny CNN) that continuously estimates the residual signal's kurtosis ("peakiness"), classifies the noise as stationary, non-stationary, or impulsive, and automatically adjusts γ and the threshold-offset parameter β . This is the direct AI/ML "adaptive" component the problem statement asks for.

5.3 Innovation 2 — Close the Paper's Gaussian/Stationary-Noise Gap

The paper openly admits weak performance on steady Gaussian noise. Add a small ML-based mask estimator trained specifically on defence-relevant stationary noise (vehicle engines, HVAC, generators) to strengthen exactly this acknowledged weakness.

5.4 Innovation 3 — Speech-Intelligibility Guard

Retain and formalise the paper's spectral-correction step (removing regained low-frequency "rumble" energy after interpolation) as a dedicated safeguard module, ensuring cleaning never compromises voice/command clarity.

5.5 Innovation 4 — Embedded/Real-Time Optimisation

Reuse the paper's low-complexity design choices (small block size, low AR order, single-pass interpolation) as hard design constraints, and benchmark AR order / block length / γ against latency and CPU budget on the target embedded platform (e.g. ARM Cortex-M/DSP, Raspberry Pi, FPGA).

5.6 Innovation 5 — Rigorous, Paper-Grounded Evaluation

Adopt the paper's own evaluation protocol — SNR improvement, PEMO/PESQ-style perceptual quality — and extend it with **STOI** (Short-Time Objective Intelligibility) to directly satisfy the problem statement's speech-intelligibility requirement.

5.7 Innovation 6 — Operator-Selectable Mission Modes

Since γ has a known, predictable relationship to noise reduction (per the paper's analytic model), expose it as a switchable mode: a **Stealth/Listening Mode** (low γ , preserves ambient situational cues) and a **Combat/Comms Mode** (high γ , aggressive cleaning to keep voice commands clear during gunfire or explosions).

6. Summary Pitch

"We take a proven, lightweight, two-stage noise-reduction architecture (frequency masking + AR interpolation) that has already been shown to outperform standard methods on spiky/impulsive noise while protecting the real signal — and we make it truly adaptive by adding a small ML classifier that automatically tunes its cleaning strength based on detected noise type, plus a stationary-noise module to cover its one known weakness — all light enough to run in real time on embedded defence hardware."

Source paper: Ruhland, M., Bitzer, J., Brandt, M., & Goetze, S. (2015). Reduction of Gaussian, Supergaussian, and Impulsive Noise by Interpolation of the Binary Mask Residual. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 23(10), 1680–1691.